



KILLARA HIGH SCHOOL

Year 12, 2 Unit Advanced Mathematics Half-Yearly Examination

Reading time: 5 mins

Examination: 3 hours

Section I

Questions 1 to 10 – Multiple Choice (10 marks , 1 mark per question)

Allow 18 minutes for this section of the paper

Section II

Questions 11 to 16 – 6 questions of 15 marks each(90 marks).

Allow 27 minutes for each of the 6 questions in this section of the paper.

Each question must be answered in a separate writing booklet.

Instructions:

- Approved calculators may be used
- Answer Multiple choice questions on provided answer sheet (at back of this test)
- Standard Integral sheet is provided (at back of this test)
- You should use a separate booklet for each of questions 11 to 16
- Show all of your working clearly.
- Marks will only be awarded to answers that are recorded on the Multiple Choice Answer sheets, or the answer booklets. You will receive no marks for working on this examination paper.

Section I

Multiple Choice

1. Evaluate $\sqrt{\frac{235.1-4.15}{4+1.3}}$ correct to two significant figures
 - A. 15.3
 - B. 6.6
 - C. 7.68
 - D. 2.87

2. Factorise $x^3 - 27$
 - A. $(x - 3)(x^2 + 6x + 9)$
 - B. $(x + 3)(x^2 - 6x + 9)$
 - C. $(x + 3)(x^2 - 3x + 9)$
 - D. $(x - 3)(x^2 + 3x + 9)$

3. $(x - 4)^2 + (y + 1)^2 = 16$ is a
 - A. Parabola, focus (4, -1)
 - B. Circle, radius = 16, centre (4, -1)
 - C. Circle, radius = 4, centre (-4, 1)
 - D. Circle, radius = 4, centre (4, -1)

4. $\int x^5 + x^3 + 1 \, dx$ is equal to
 - A. $\frac{x^6}{6} + \frac{x^4}{4} + x + C$
 - B. $\frac{x^6}{6} + \frac{x^4}{4} + x$
 - C. $x^6 + x^4 + x + C$
 - D. $x^6 + x^4 + x$

5. $\frac{d}{dx} (x \cdot \sin x)$ is equal to
- A. $\cos^2 x$
 - B. $x \cdot \sin x + \cos x$
 - C. $x \cdot \cos x + \sin x$
 - D. $\sin^2 x$
6. $\int \sin \frac{x}{4} dx$ is equal to
- A. $\frac{1}{4} \sin \frac{x}{4} + C$
 - B. $-\frac{1}{4} \cos \frac{x}{4} + C$
 - C. $4 \sin \frac{x}{4} + C$
 - D. $-4 \cos \frac{x}{4} + C$
7. The curve $y = 4 \cos 3x$ has
- A. Amplitude 4 and Period $\frac{2\pi}{3}$
 - B. Amplitude 4 and Period $\frac{\pi}{3}$
 - C. Amplitude 3 and Period $\frac{\pi}{2}$
 - D. Amplitude 3 and Period 4π
8. The curve $y = f(x)$ has gradient function $\frac{dy}{dx} = 3x^2 + 4x - 1$ and passes through the point B(0,1) . Its equation is given by
- A. $y = x^3 + 2x^2 - x + C$
 - B. $y = x^3 + 2x^2 - x + 2$
 - C. $y = x^3 + 2x^2 - x + 1$
 - D. $y = 6x + 4$

9. The point of inflexion of the curve $y = x^3 - 12x + 1$ is

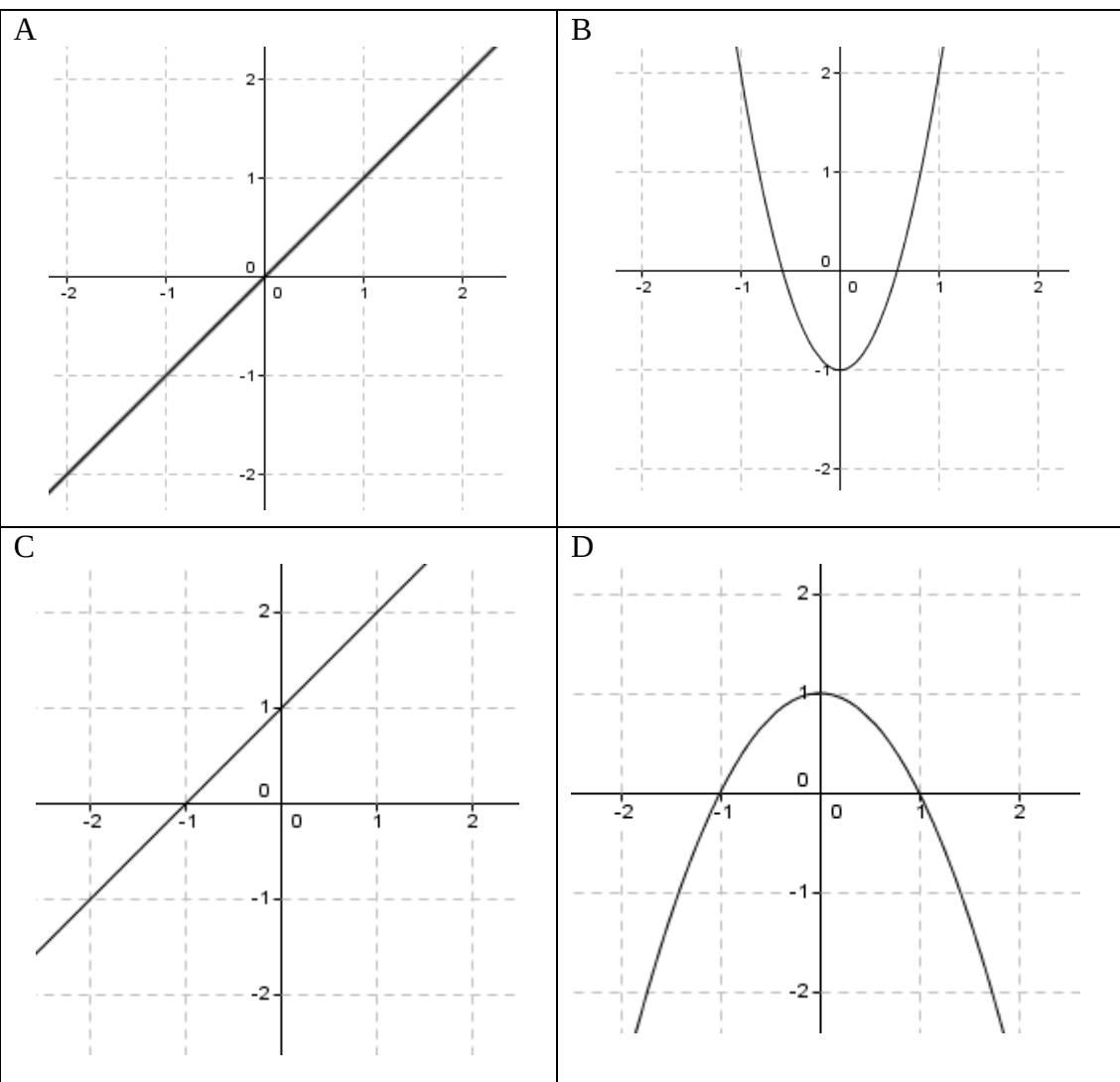
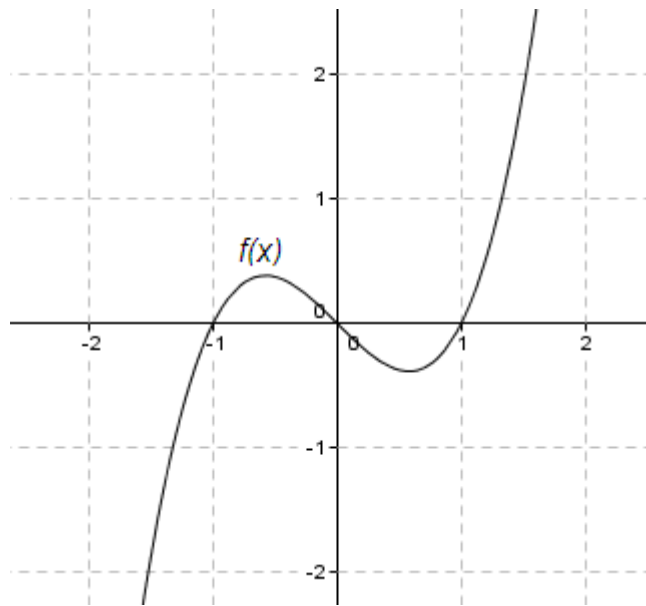
A. (2 , -15)

B. (-2 , 17)

C. (0 , 0)

D. (0 , 1)

10. The curve $f(x)$ is shown below, the graph of its gradient function is



Section II

please answer each question in a new answer booklet

Marks

11. (15 marks)

- a) For the equation $5x^2 + 2x + k = 0$
- i. What is the discriminant? 1
 - ii. For what values of k does $5x^2 + 2x + k = 0$ have real roots? 1
- b) Using the substitution $u = x^2$ solve the equation $4x^4 - 5x^2 - 9 = 0$ 2
- c) Find the coordinates of the focus of the parabola with equation $x^2 = 4(y - 1)$. 1
- d) Express $(2\sqrt{3} + 1)(2 - \sqrt{3})$ in the form $a\sqrt{3} + b$ 2
- e) Solve the inequality $|3x - 5| \geq 2$ 2
- f) Given the points A(1,3) and B(4,-2)
- i. Determine the equation of the line AB 1
 - ii. What is the size of the acute angle that AB makes with the x axis (to the nearest degree)? 1
- g) Prove $\frac{\cot\theta}{\operatorname{cosec}\theta - 1} - \frac{\cos\theta}{1 + \sin\theta} = 2 \tan\theta$ 2
- h) If $\sin\theta = -\frac{4}{11}$ and $\tan\theta > 0$ find the exact value of $\cos\theta$ 2

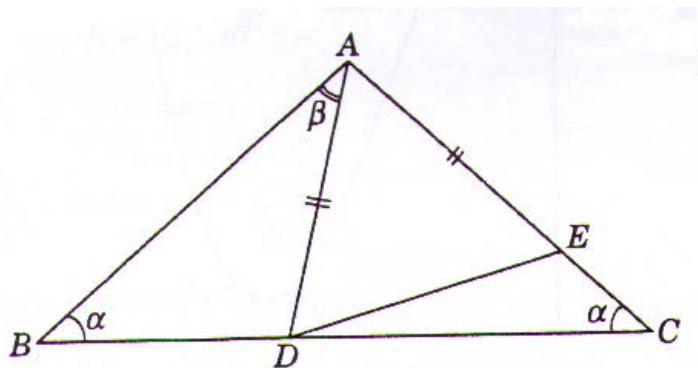
please answer each question in a new answer booklet

Marks

12. (15 marks)

a) In the isosceles triangle ABC as shown below, $\angle ABC = \angle ACB = \alpha$.

The points D and E lie on BC and AC , so that $AD = AE$, as shown in the diagram. Let $\angle BAD = \beta$



i. Explain why $\angle ADC = \alpha + \beta$

1

ii. Find $\angle DAC$ in terms of α and β

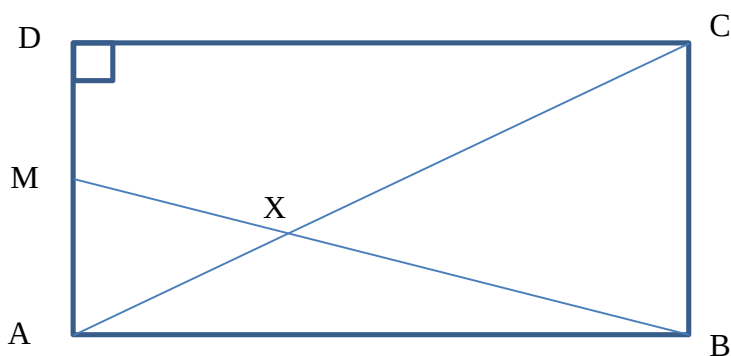
1

iii. Hence, or otherwise, find $\angle EDC$ in terms of β

2

b) In the diagram below $ABCD$ is a rectangle and $AB = 2AD$.

The point M is the midpoint of AD . The line BM meets AC at X .



i. Show that $\triangle AXM \sim \triangle BXC$

2

ii. Show that $3CX = 2AC$

2

Marks

c) Differentiate $x^2 + x$ from first principles

2

d) Find the equation of the tangent to the curve

2

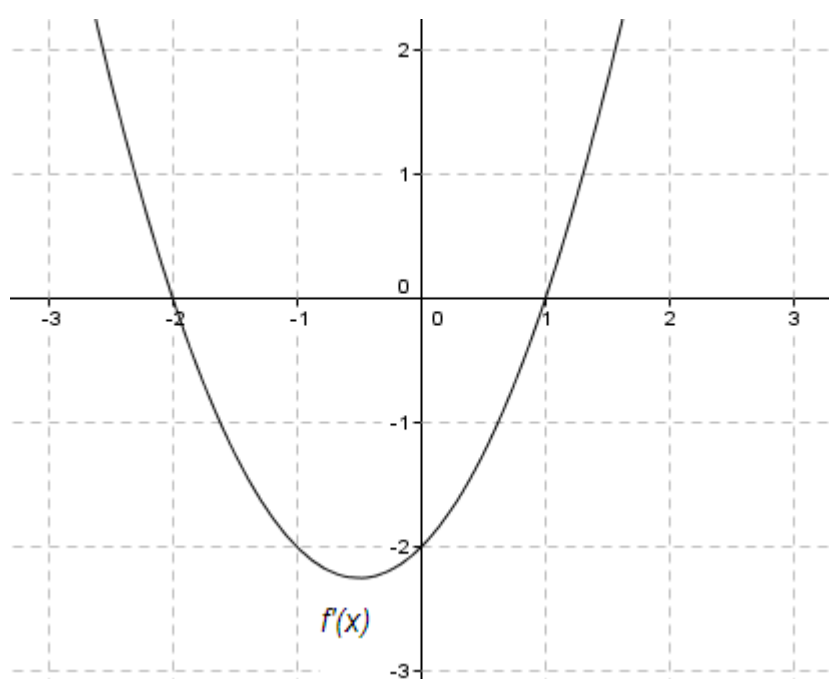
$$y = x^2 - 3x + 5 \text{ when } x = 1$$

e) Differentiate $\frac{\sin x}{(x+1)^2}$

2

f) Sketch a primitive function $f(x)$ given the graph of the derivative function $f'(x)$ as shown

1



please answer each question in a new answer booklet

Marks

13. (15 marks)

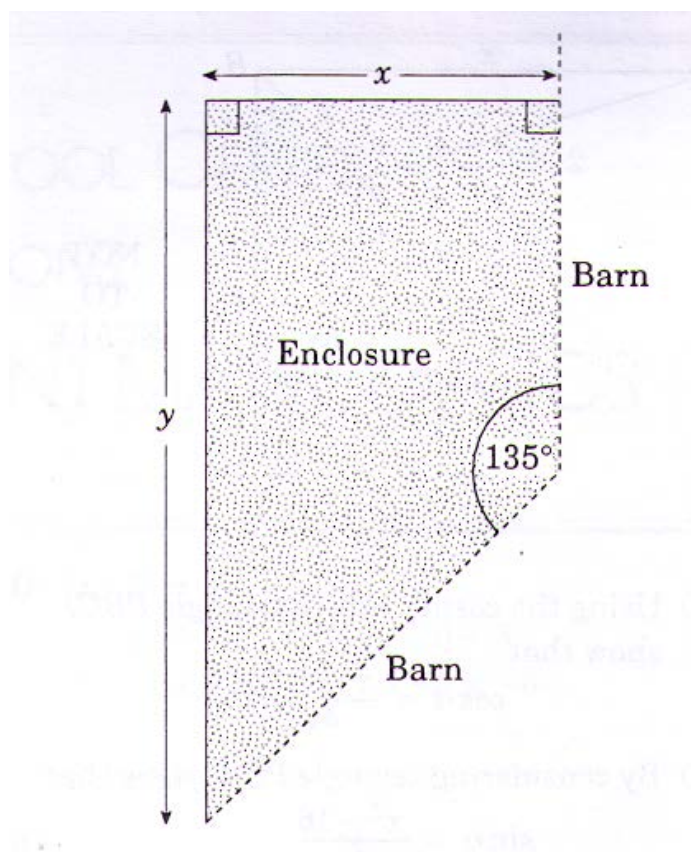
- a) Consider the curve given by $y = \frac{1}{3}x^3 - x^2$
- i. Find any turning points and determine their nature 3
 - ii. Find any points of inflexion 1
 - iii. Sketch the curve in the domain $-1 \leq x \leq 4$ clearly 2
indicating all important points.
 - iv. For what values of x is the curve concave down? 1
- b) The curve $y = 3x^2 - \frac{p}{x^2}$ has a turning point at $x = -1$. 2
- Find the constant p .

- c) An enclosure is to be built adjoining a barn, as in the diagram Marks
- shown below. The walls of the barn meet at 135° , and 117 metres of fencing is available for the enclosure, so that $x + y = 117$ where x and y are as shown in the diagram.

- i. Show that the shaded area of the enclosure is given by 3

$$A = 117x - \frac{3}{2}x^2$$

- ii. Show that the largest area of the enclosure occurs when 3
- $y = 2x$.



please answer each question in a new answer booklet

Marks

14. (15 marks)

a) Consider the function $f(x) = x^4 - x$.

- i. Copy the table of values below into your writing booklet and calculate the missing values.

x	0	1	2	3	4
y	0				252

1

- ii. Use these five values and the Trapezoidal rule to approximate

2

the value of $\int_0^4 x^4 - x \, dx$

- iii. Use these five values and Simpson's rule to approximate the

2

value of $\int_0^4 x^4 - x \, dx$

- iv. Calculate the exact value of $\int_0^4 x^4 - x \, dx$

2

- v. Place in ascending order of accuracy your results for Simpson's rule, exact value and the Trapezoidal rule

1

- vi. Briefly justify the ordering of your results in part v.

1

b) Considering the equations of $y = 2x$ and $y = \frac{1}{2}x^3$

- i. Find the coordinates of the points of intersection

2

- ii. Find the area between the 2 curves.

2

- iii. Find the volume of the solid of revolution formed by rotating about the x-axis the area between the curves that is between the lines $x=2$ and $x=0$.

2

please answer each question in a new answer booklet

Marks

15. (15 marks)

- a) Sketch the curve $y = 1 - \cos 2x$ for $0 \leq x \leq \pi$

3

Clearly show points at $x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4},$ and π

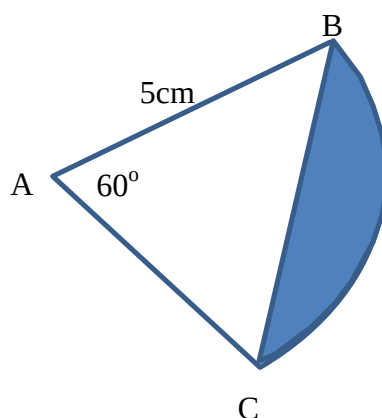
- b) Solve $2\sin^2 x - 3\sin x - 2 = 0$ for $0 \leq x \leq 2\pi$

2

- c) Calculate $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$

2

- d) For the sector of the circle shown below

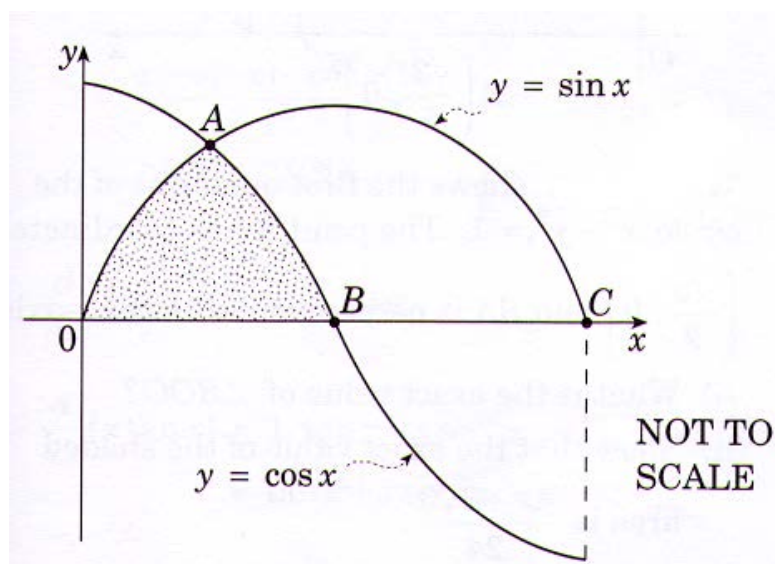


- i. Calculate the exact arc length of the sector
- ii. Calculate the exact Area of the shaded minor segment

1

3

- e) The diagram below shows part of the curve of $y = \sin x$ and $y = \cos x$



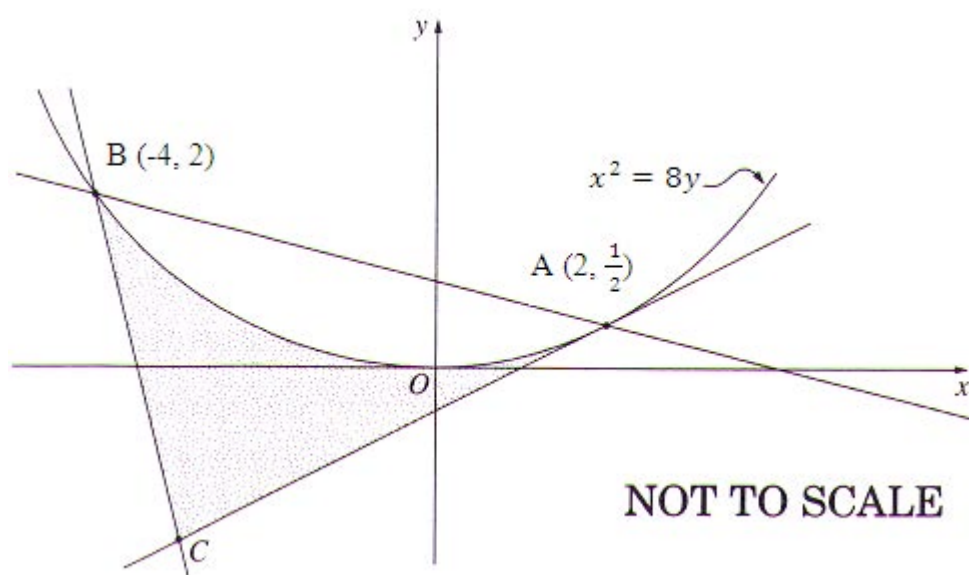
- i. Find the coordinates of the points A and B
- ii. Calculate the Area of the shaded region

2

2

16. (15 marks)

- a) The diagram below shows the graph of the parabola $x^2 = 8y$. The points A $(2, \frac{1}{2})$ and B $(-4, 2)$ are on the parabola, and C is the point where the tangent to the parabola at A intersects the directrix.



- i. Write down the equation of the directrix of the parabola $x^2 = 8y$ 1
- ii. Show that the equation of the tangent to the parabola at point A is the line $x - 2y - 1 = 0$ 2
- iii. Show that C is the point $(-3, -2)$. 1
- iv. Given that the equation of the line AB is $y = 1 - \frac{x}{4}$, find the area bounded by the line AB and the parabola. 2
- v. Hence, or otherwise find the shaded Area bounded by the parabola, the tangent at A and the line BC. 3

b)	Marks
i. Show that $x = \frac{\pi}{6}$ is a solution of $\cot x = \tan 2x$	1
ii. On the same set of axes, sketch the graphs of the functions $y = \cot x$ and $y = \tan 2x$ for $-\pi \leq x \leq \pi$. Use a different pen colour, or different style of line for $y = \cot x$ and $y = \tan 2x$.	3
iii. Hence find the number of solutions of $\cot x = \tan 2x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$	1
iv. Use your graphs to solve $\cot x \leq \tan 2x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$	1

END OF EXAMINATION

2014 McCartney Q 13
~~2013~~ - Killara - 2 Unit Half Yearly
 Solutions

Section 1

1 B

2 D

3 D

4 A

5 C

6 D

7 A

8 C

9 D

10 B

Worked solutions to Section 1

$$1 \quad \sqrt{\frac{(235.1 - 4.5)}{(4 + 1.3)}} = \sqrt{\frac{230.6}{5.3}}$$

$$= 6.60117...$$

$$= 6.6 \text{ 2 sig fig}$$

$$= \textcircled{B} \text{ option}$$

$$2 \quad x^3 - 27$$

$$= x^3 - 3^3$$

$$= (x-3)(x^2 + 3x + 9)$$

$$= \textcircled{D} \text{ option}$$

$$3 \quad (x-4)^2 + (y+1)^2 = 16$$

$$= (x-4)^2 + (y+1)^2 = 4^2$$

circle centre (4, -1), r = 4

$$= \text{option } \textcircled{D}$$

$$4 \quad \int x^5 + x^3 + 1 \, dx$$

$$= \frac{x^6}{6} + \frac{x^4}{4} + x + C$$

$$= \text{option } \textcircled{A}$$

$$5 \quad y = x \cdot \sin x$$

$$y' = u'v + v'u$$

$$y' = 1 \cdot \sin x + \cos x \cdot x$$

$$y' = x \cdot \cos x + \sin x$$

$$= \text{option } \textcircled{C}$$

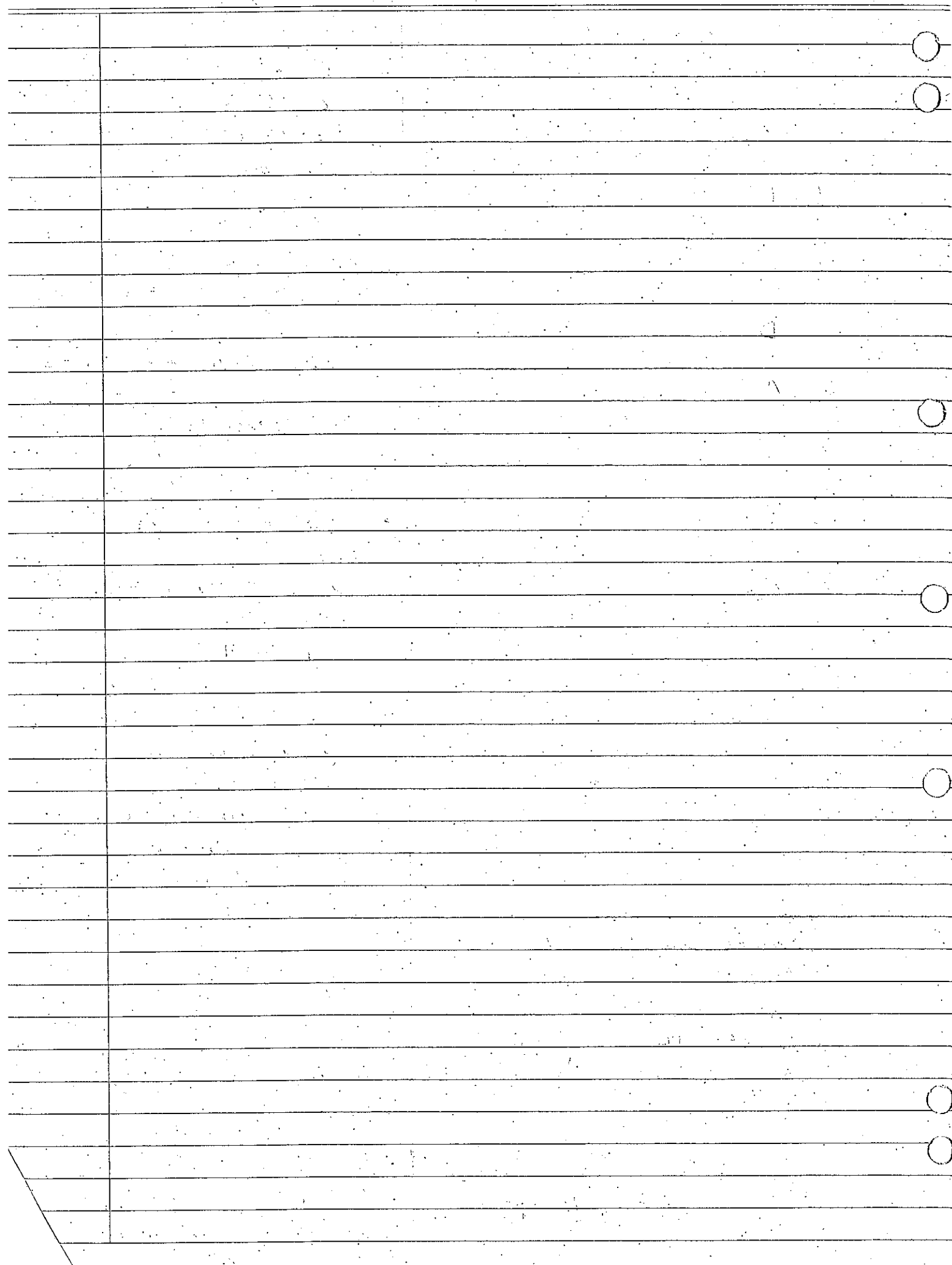
$$6 \quad \int \sin \frac{x}{4} \, dx$$

we know if $y = -\cos \frac{x}{4}$

$$y' = \frac{1}{4} \sin \frac{x}{4}$$

we need -4, out the front

$$\cos \frac{x}{4} \quad \therefore y = -4 \cos \frac{x}{4} + C$$



Section 1 answers

continued...

7 $y = 4 \cos 3x$

↑
amplitude

↑
period = $2\pi \div 3$
 $= \frac{2\pi}{3}$

$\therefore$ option (A)

8 $y' = 3x^2 + 4x - 1$
 $y = x^3 + 2x^2 - x + C$

sub (0, 1)

$$1 = 0^3 + 2 \times 0^2 - 0 + C$$

$$\therefore C = 1$$

$$\therefore y = x^3 + 2x^2 - x + 1$$

$\therefore$ option (C)

9 $y = x^3 - 12x + 1$

$$y' = 3x^2 - 12$$

$$y'' = 6x$$

when $y'' = 0$, $x = 0$

sub $x = 0$ into

$$y = x^3 - 12x + 1$$

$$y = 0^3 - 12 \times 0 + 1$$

$$\therefore y = 1$$

$\therefore$ point is (0, 1)

check a point of inflexion.

at $x = -1$, $y'' = -6$

$x = 1$, $y'' = 6$

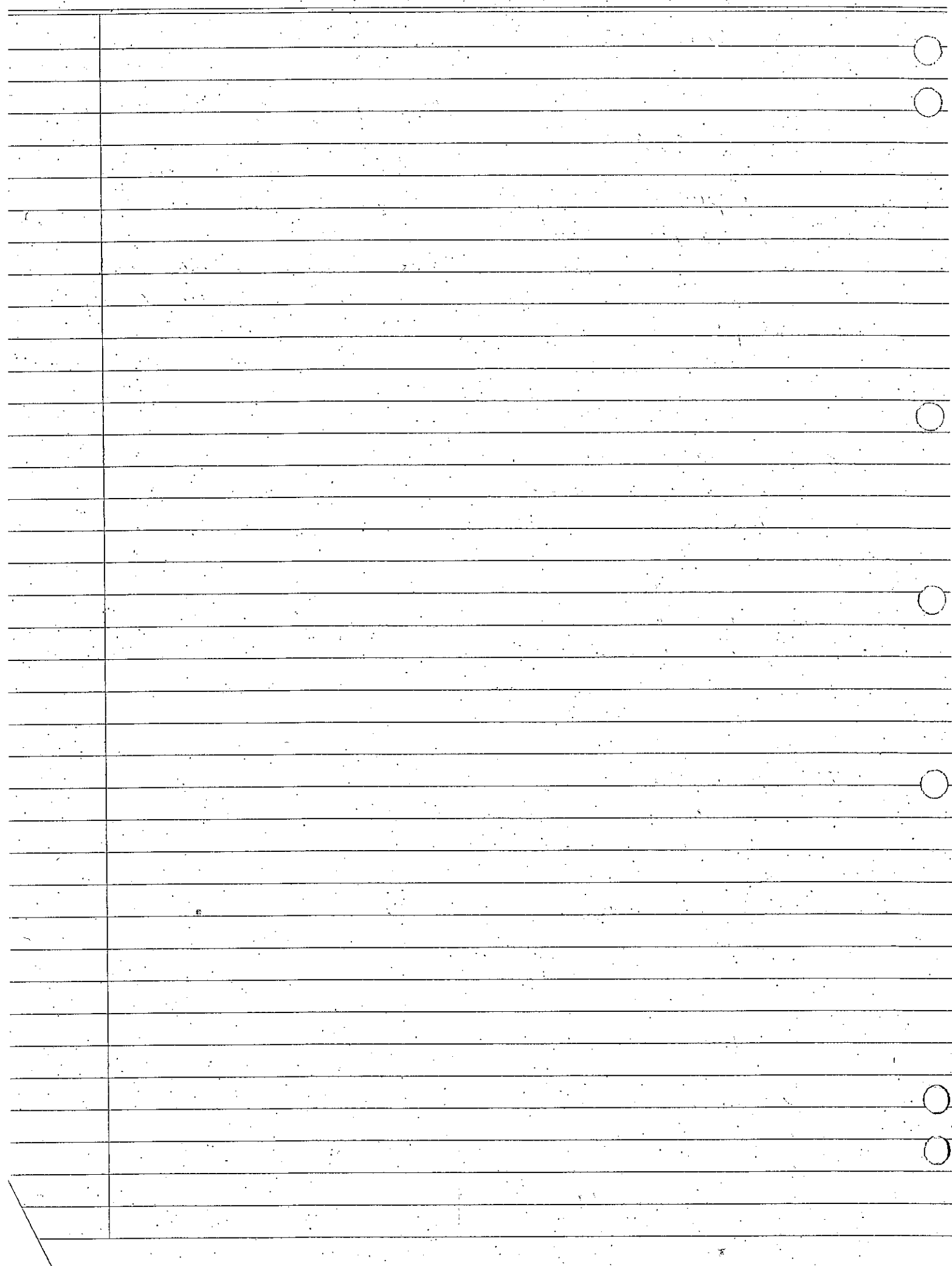
$\therefore$ (0, 1) is point of inflexion

$\therefore$ option (D)

10 option B is the answer

Note: crosses x-axis
in graph B at $x = \pm \frac{1}{2}$

where turning points are
on $f(x)$ i.e. gradient = 0.



Q11

a i

$$\Delta = b^2 - 4ac$$

$$\Delta = 2^2 - 4 \times 5 \times k$$

$$\Delta = 4 - 20k$$

$$\Delta = 4(1 - 5k)$$

ii when $\Delta \geq 0$ we have real roots

$$1 - 5k \geq 0, -5k \geq -1$$

$$k \leq \frac{1}{5}$$

b

$$4x^4 - 5x^2 - 9 = 0$$

$$\text{let } u = x^2$$

$$4u^2 - 5u - 9 = 0$$

$$(4u - 9)(u + 1) = 0$$

$$\therefore u = \frac{9}{4}, u = -1$$

(1 mark for solving correctly for u)

$$\text{when } u = \frac{9}{4}, x^2 = \frac{9}{4}$$

$$x = \pm \frac{3}{2}$$

$$\text{when } u = -1, x^2 = -1$$

$\therefore$ no real values for x

$\therefore$ Solutions are $x = \pm \frac{3}{2}$ (1 mark for correct solution, or ECF)

solution from the u values)

c

$$x^2 = 4(y - 1)$$

$$\text{vertex} = (0, 1)$$

$$x^2 = 4a(y - 1) \quad \therefore a = 1$$

$$\therefore \text{Focus} = (0, 2)$$

d $(2\sqrt{3}+1)(2-\sqrt{3})$

$$= 4\sqrt{3} - 6 + 2 - \sqrt{3}$$

$$= 3\sqrt{3} - 4$$

in form $a\sqrt{3} + b$

$$\therefore a=3, b=-4$$

(1 mark for a)

(1 mark for b)

e $|3x-5| > 2$

$$3x-5 < -2$$

or

$$3x-5 > 2$$

$$3x < 3$$

$$3x > 7$$

$$x < 1$$

$$x > \frac{7}{3}$$

(1 mark)

(1 mark)

f $A(1,3); B(4,-2)$

i gradient of AB $= \frac{3-(-2)}{1-4}$
 $= -\frac{5}{3}$

ii equation of AB

$$y-y_1 = m(x-x_1)$$

$$y-3 = -\frac{5}{3}(x-1)$$

$$3y-9 = -5x+5$$

$$5x+3y-14=0$$

iii obtuse angle $= 180^\circ - \tan^{-1}\left(\frac{5}{3}\right)$
 $= 180^\circ - 59^\circ$
 $= 121^\circ$

$$9 \quad LHS = \frac{\cot \theta}{\operatorname{cosec} \theta - 1} - \frac{\cot \theta}{1 + \sin \theta}$$

$$= \frac{\cot \theta}{1 - \sin \theta} - \frac{\cot \theta}{1 + \sin \theta}$$

$$\therefore \operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$= \frac{\cot \theta (1 + \sin \theta) - \cot \theta (1 - \sin \theta)}{1 - \sin^2 \theta}$$

$$= \frac{\cot \theta (2 \sin \theta)}{1 - \sin^2 \theta} \quad (1 \text{ mark})$$

$$= \frac{\cancel{\cot \theta} \cdot 2 \sin \theta}{\cancel{\cot \theta}}$$

$$\therefore \sin^2 \theta + \cos^2 \theta = 1$$

$$= \frac{2 \sin \theta}{\cot \theta} \quad (1 \text{ mark})$$

$$= 2 \tan \theta = RHS.$$

OR alternatively ---

$$LHS \frac{\cot \theta}{\operatorname{cosec} \theta - 1} \times \frac{\operatorname{cosec} \theta + 1}{\operatorname{cosec} \theta + 1} - \frac{\cot \theta}{1 + \sin \theta} \times \frac{1 - \sin \theta}{1 - \sin \theta}$$

$$= \frac{\cot \theta (\operatorname{cosec} \theta + 1)}{\operatorname{cosec}^2 \theta - 1} + \frac{-\cot \theta (1 - \sin \theta)}{1 - \sin^2 \theta}$$

$$(1 \text{ mark}) \quad = \frac{\cot \theta (\operatorname{cosec} \theta + 1)}{\cot^2 \theta} + \frac{-\cot \theta (1 - \sin \theta)}{\cot^2 \theta}$$

$$\therefore 1 - \cot^2 \theta = \operatorname{cosec}^2 \theta$$

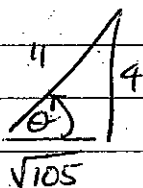
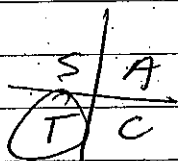
$$= \frac{\operatorname{cosec} \theta}{\cot \theta} + \frac{1}{\cot \theta} - \frac{1}{\cot \theta} + \frac{\sin \theta}{\cot \theta}$$

$$1 - \sin^2 \theta = \cos^2 \theta$$

$$= \frac{1}{\sin \theta} \times \frac{\sin \theta}{\cos \theta} - \frac{1}{\cos \theta} + \tan \theta + \tan \theta \quad (1 \text{ mark})$$

$$= 2 \tan \theta = RHS.$$

h $\sin \theta = \frac{-4}{11}$ and $\tan \theta > 0$



$$\pi < \theta < \frac{3\pi}{2}$$

$$\cos \theta < 0$$

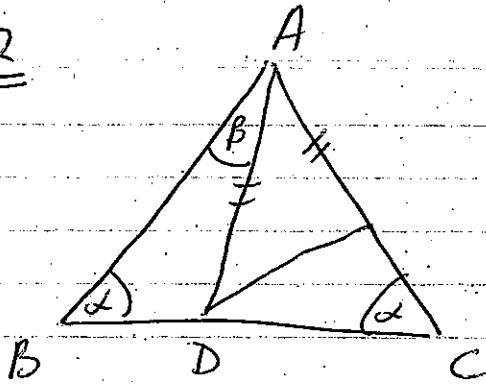
$$\therefore \cos \theta = -\frac{\sqrt{105}}{11}$$

(1 mark quadrant
 & sign).

(1 mark positive or negative
 version of $\frac{\sqrt{105}}{11}$)

Q12

a



i in $\triangle ABD$

ext $\angle ADC = \text{sum of the interior opposite } \angle\text{'s}$
 $= \alpha + \beta$

ii in $\triangle ADC$

$\angle ADC + \angle DCA + \angle DAC = 180^\circ$ (sum of angles in a $\triangle$)

$$(\alpha + \beta) + \alpha + \angle DAC = 180^\circ$$

$$\angle DAC = 180^\circ - (2\alpha + \beta)$$

iii

In $\triangle ADE$

$$\angle DAC = 180^\circ - (2\alpha + \beta) \text{ from ii}$$

$$\angle ADE = \frac{180 - \angle DAC}{2}$$

$$= \frac{180 - [180 - (2\alpha + \beta)]}{2}$$

$$= \frac{\alpha + \beta}{2}$$

$$\angle ADE + \angle EDC = \angle ADC = \alpha + \beta$$

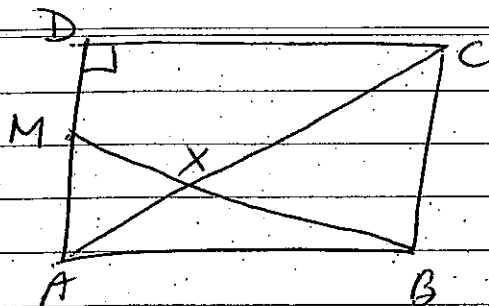
$$\frac{\alpha + \beta}{2} + \angle EDC = \alpha + \beta$$

$$\therefore \angle EDC = \frac{\beta}{2}$$

(1 mark)

validly show
(1 mark) $\angle ADE = \frac{\alpha + \beta}{2}$
or similar
step to
answer
makes a conclusion
and gets
 $\angle EDC$.

12b



i In $\triangle AXM$ and $\triangle BXC$

(1 mark body)

$\angle AXM = \angle BXC$ (vertically opposite $\angle$'s)

$\angle XMA = \angle XCB$ (alternate $\angle$'s of parallel lines, $AD \parallel BC$).

(1 mark conclusion)

$\therefore \triangle AXM \parallel \triangle BXC$ (Equiangular)

ii $\frac{AX}{CX} = \frac{AM}{BC}$ (matching sides of similar triangles).

$$\frac{AX}{CX} + 1 = \frac{AM}{BC} + 1$$

$$\frac{AX+CX}{CX} = \frac{AM+BC}{BC}$$

$$\therefore BC = 2AM$$

$$AC = AX + XC$$

$$\frac{AC}{CX} = \frac{AM+2AM}{2AM}$$

$$\frac{AC}{CX} = \frac{3}{2}$$

$$\therefore 3CX = 2AC$$

(1 mark) correctly uses ratios to get answer.

OR

$$CX : AC$$

$$CX : AX + XC$$

$$2 : 1+2 \quad \therefore AC = AX + XC$$

and as $BC : AM = 2 : 1$ as M is midpoint

other sides in ratio 2:1 of similar Δ 's

$$\therefore CX : AC = 2 : 3$$

$$\frac{CX}{AC} = \frac{2}{3} \quad \text{cross multiply}$$

$$3CX = 2AC \quad (1 \text{ mark})$$

12c Let $f(x) = x^2 + x$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

← must show

$$f'(x) = \lim_{h \rightarrow 0} \frac{[(x+h)^2 + x+h] - [x^2 + x]}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + x + h - x^2 - x}{h} \quad (1 \text{ mark})$$

$$f'(x) = \frac{h(2x+1+h)}{h} \lim_{h \rightarrow 0}$$

$$f'(x) = 2x+1$$

(1 mark)
final answer.

Note 1 mark given out of 2
for good process but errors made

d

$$y = x^2 - 3x + 5$$

$$y' = 2x - 3$$

$$m_T = 2(1) - 3 = -1 \text{ at } x=1$$

$$\text{when } x=1, y = 1 - 3 + 5 = 3$$

$$\therefore m_T = -1, \text{ point is } (1, 3) \quad (1 \text{ mark})$$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = -1(x - 1)$$

$$y - 3 = -x + 1$$

$$x + y - 4 = 0 \quad (1 \text{ mark})$$

or if correct equation
for their incorrect
gradient.

$$e \quad y = \frac{\sin x}{(x+1)^2}$$

$$\begin{cases} y = \frac{u}{v} \\ y' = \frac{vu' - uv'}{v^2} \end{cases}$$

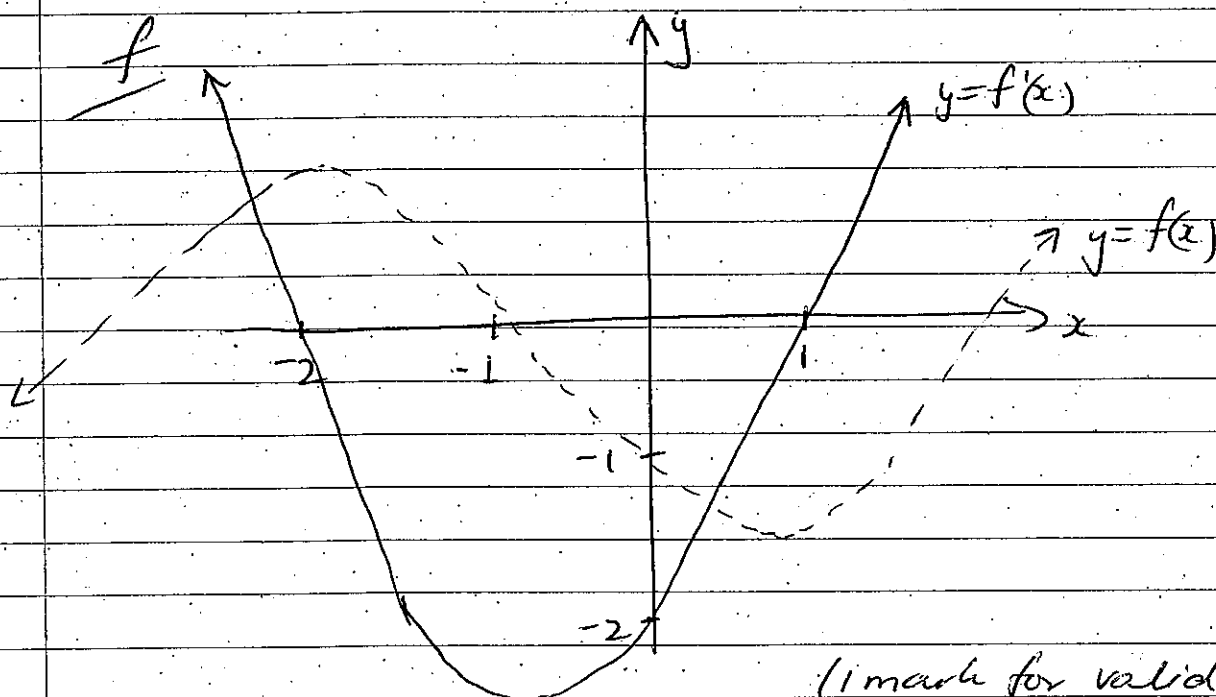
$$y' = \frac{(x+1)^2 \cdot \cos x - 2(x+1) \cdot \sin x}{(x+1)^4}$$

(1 mark)
substitutes
into quotient
rule

$$y' = \frac{[(x+1) \cdot \cos x - 2 \sin x] \cdot \cancel{(x+1)}}{(x+1)^3}$$

$$\therefore y' = \frac{(x+1) \cdot \cos x - 2 \sin x}{(x+1)^3}$$

(1 mark)
final answer.



(1 mark for valid
attempt,
must have
turning points
at -2 and 1

13. a $y = \frac{1}{3}x^3 - x^2$

i $y' = x^2 - 2x$

$y' = x(x-2)$

$\therefore y' = 0$ when $x=0, x=2$

~~(1 mark)~~ $\therefore$ stationary points are $(0,0)$ and $(2, -\frac{4}{3})$

$y'' = 2x - 2$

~~(1 mark for max)~~ when $x=0$ $\frac{d^2y}{dx^2} < 0 \therefore (0,0)$ is a maximum turning point

ECF 0.6
~~(1 mark for min)~~ when $x=2$ $\frac{d^2y}{dx^2} > 0 \therefore (2, -\frac{4}{3})$ is a minimum turning point

Note: must show actual values
if y' used to determine min & max.

3

ii when $y'' = 0$ $y'' = 2(x-1)$
 $x=1$

x	0.9	1	1.1
y''	-0.2	0	+0.2

$\therefore$ change in concavity at $x=1$
must show change in concavity

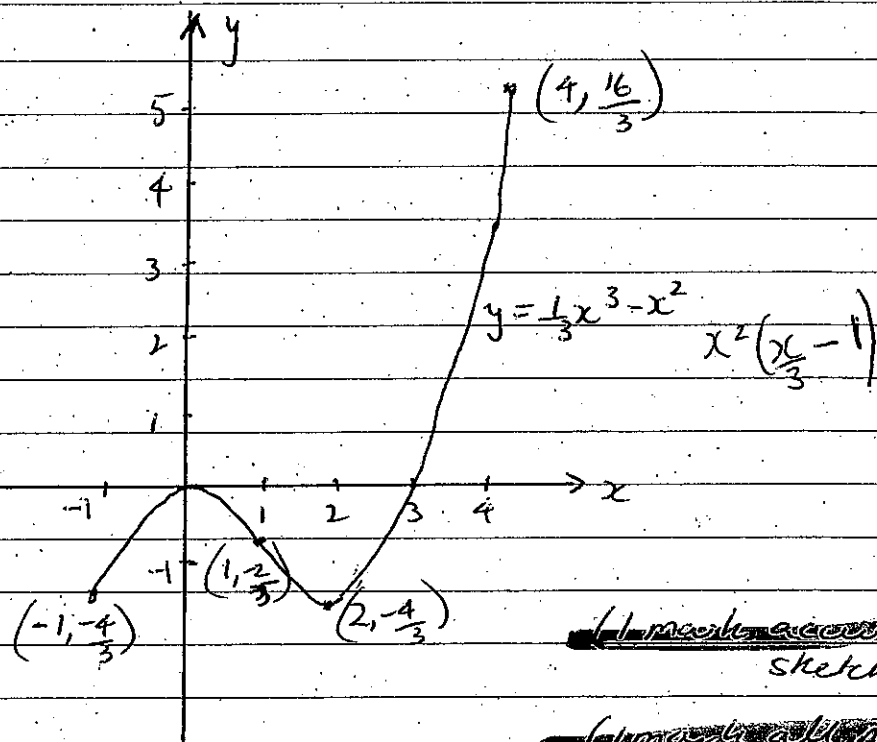
~~$\therefore (1, -\frac{2}{3})$ is a point of inflexion~~

1

W

Y

iii



(1 mark) accurate sketch

(1 mark) all points indicated

2
-1 if
error or
some info missing

iv curve is concave down
for $-1 < x < 1$ (from the graph)
 $x < 1$ ok. ~~must have down restriction~~
 $x < 1$ is not.

13b $y = 3x^2 - \frac{p}{x^2}$

$y' = 6x + \frac{2p}{x^3}$ (1 mark) for first derivative

when $x = -1, y' = 0$

$\therefore 0 = 6x - 1 + \frac{2p}{-1}$

$0 = -6 - 2p$

$2p = -6$

$(0 = -6 - 2p) \times -1$

$p = -3$

$2p + 6 = 0$

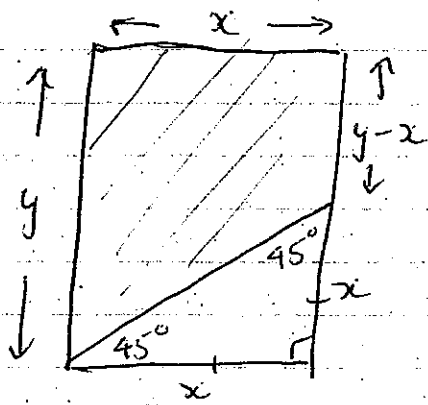
$2p = -6$

$\therefore p = -3$

(1 mark) final answer

2

130
i



$$x+y=117$$

Area of unshaded Δ

$$= \frac{1}{2} x^2 \quad (1 \text{ mark})$$

Area of $\square = xy$

$$= x(117-x) \quad (1 \text{ mark})$$

$$A = xy - \frac{1}{2} x^2$$

$$A = x(117-x) - \frac{1}{2} x^2$$

$$A = 117x - \frac{3}{2} x^2 \quad (1 \text{ mark})$$

3

ii

$$\frac{dA}{dx} = 117 - 3x$$

$$\text{when } \frac{dA}{dx} = 0$$

$$\left. \begin{aligned} 0 &= 117 - 3x \\ 3x &= 117 \\ x &= 39 \end{aligned} \right\} 1 \text{ mark}$$

$$\frac{d^2A}{dx^2} = -3 \text{ which is } < 0$$

Max when $x = 39$

(1 mark)
shows maximum correctly

$$\text{when } x = 39, y = 117 - 39, y = 78$$

$$y = 2x$$

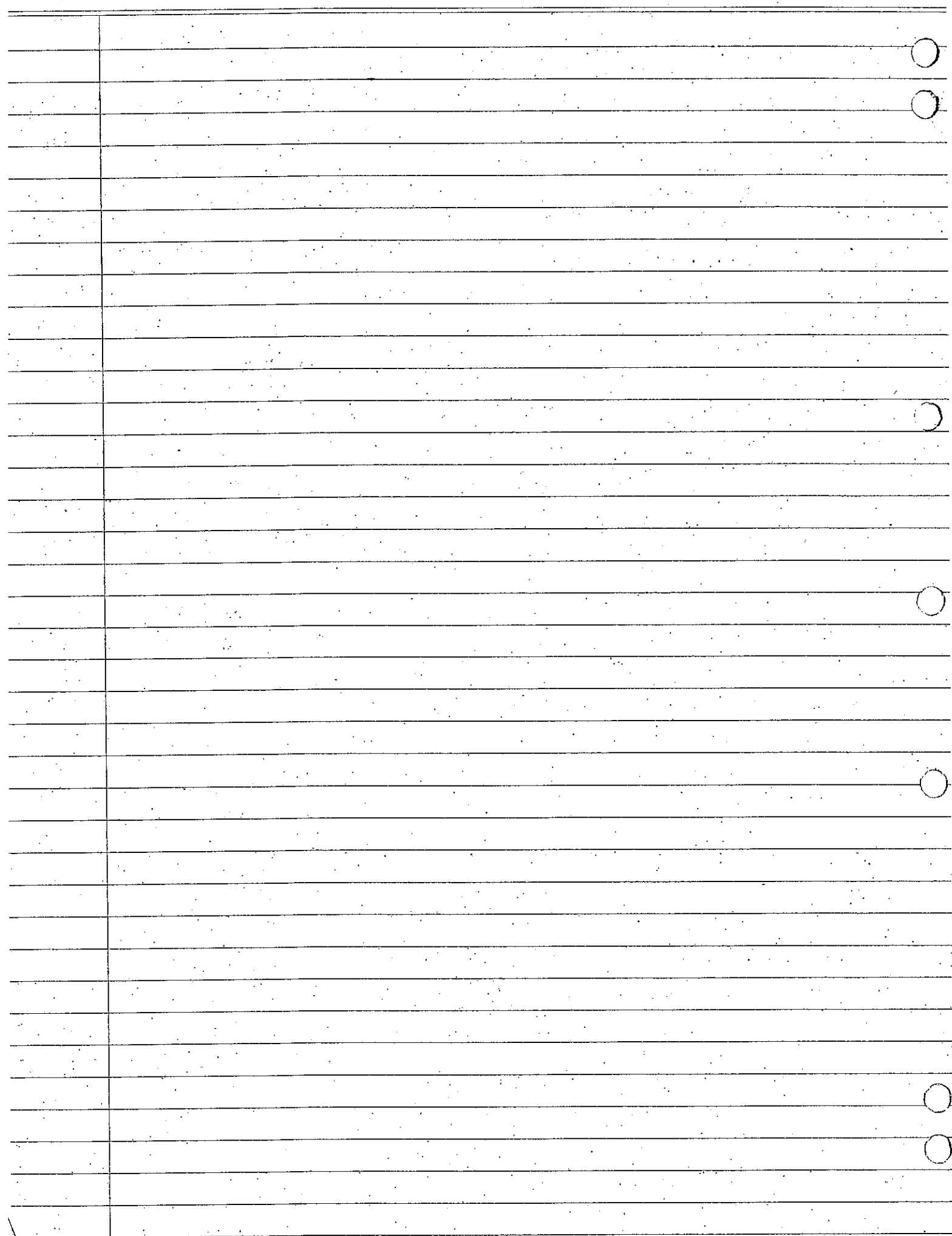
(1 mark)

shows $y = 2x$ with good reasoning

$$\therefore x = 39, y = 78$$

3

w



Q14 a $f(x) = x^4 - x$

i

x	0	1	2	3	4
y	0	0	14	78	252

(1 mark)

ii Trapezoidal Rule

$$\int_0^4 x^4 - x \, dx = \frac{h}{2} [y_0 + y_4 + 2(y_1 + y_2 + y_3)]$$

$$h = \frac{4-0}{4} = 1$$

(1 mark
uses correct
h value
and
substitution)

$$= \frac{1}{2} [0 + 252 + 2(0 + 14 + 78)]$$

$$= 218 \text{ units}^2 \quad (1 \text{ mark})$$

iii Simpson's Rule

$$\int_0^4 x^4 - x \, dx = \frac{h}{3} [(y_0 + y_4) + 4(y_1 + y_3) + 2y_2]$$

$$h = 1$$

(1 mark
h value
correct
and substitution)

$$= \frac{1}{3} [(0 + 252) + 4(0 + 78) + 2 \times 14]$$

$$= 197 \frac{1}{3} \text{ units}^2 \quad (1 \text{ mark})$$

iv $\int_0^4 x^4 - x \, dx$

$$= \left[\frac{x^5}{5} - \frac{x^2}{2} \right]_0^4$$

(1 mark correct
integration)

$$= \left[\frac{4^5}{5} - \frac{4^2}{2} \right] - 0$$

$$= 204.8 - 8$$

$$= 196.8 \text{ units}^2 \quad (1 \text{ mark answer})$$

v In ascending order of accuracy
trapezoidal rule, Simpson's rule, exact

vi Exact is done by integration and is most precise.

Simpson's rule approximates areas using curves
so matches this curve well.

trapezoidal rule, use multiple trapeziums
so doesn't match curve well.

b i $y = 2x, y = \frac{1}{2}x^3$

$$2x = \frac{1}{2}x^3$$

$$\frac{1}{2}x^3 - 2x = 0$$

$$x\left(\frac{1}{2}x^2 - 2\right) = 0 \quad (1 \text{ mark}) \text{ factorised correctly.}$$

$\therefore$ intersect when $x = 0$ and $\frac{1}{2}x^2 - 2 = 0$

$$\frac{1}{2}x^2 = 2$$

$$x^2 = 4$$

$$x = \pm 2$$

$\therefore$ points are $(0, 0), (2, 4), (-2, -4)$ (1 mark)

ii $A = 2 \int_0^2 2x - \frac{1}{2}x^3 dx$

$$A = 2 \left[x^2 - \frac{x^4}{8} \right]_0^2 \quad (\text{correct integration } 1 \text{ mark})$$

$$A = 2 \left[\left[4 - \frac{16}{8} \right] - 0 \right], \quad A = 2 \times 2 = 4 \text{ units}^2 \quad (1 \text{ mark}).$$

Q 14 b iii $V_1 = \pi \int_0^2 y^2 dx$ where $y=2x$

$$V_1 = \pi \int_0^2 (2x)^2 dx$$

$$V_1 = \pi \int_0^2 4x^2 dx$$

$$V_1 = \pi \left[\frac{4x^3}{3} \right]_0^2$$

$$V_1 = \pi \cdot \frac{32}{3} \text{ units}^3$$

$$V_2 = \pi \int_0^2 y^2 dx \text{ where } y = \frac{1}{2} x^3$$

$$V_2 = \pi \int_0^2 \left(\frac{1}{2} x^3 \right)^2 dx$$

$$V_2 = \pi \int_0^2 \frac{1}{4} x^6 dx$$

$$V_2 = \pi \left[\frac{x^7}{28} \right]_0^2$$

$$V_2 = \pi \cdot \frac{128}{28} = \frac{32\pi}{7}$$

$$\therefore V = V_1 - V_2$$

$$V = \pi \left(\frac{32}{3} - \frac{32}{7} \right)$$

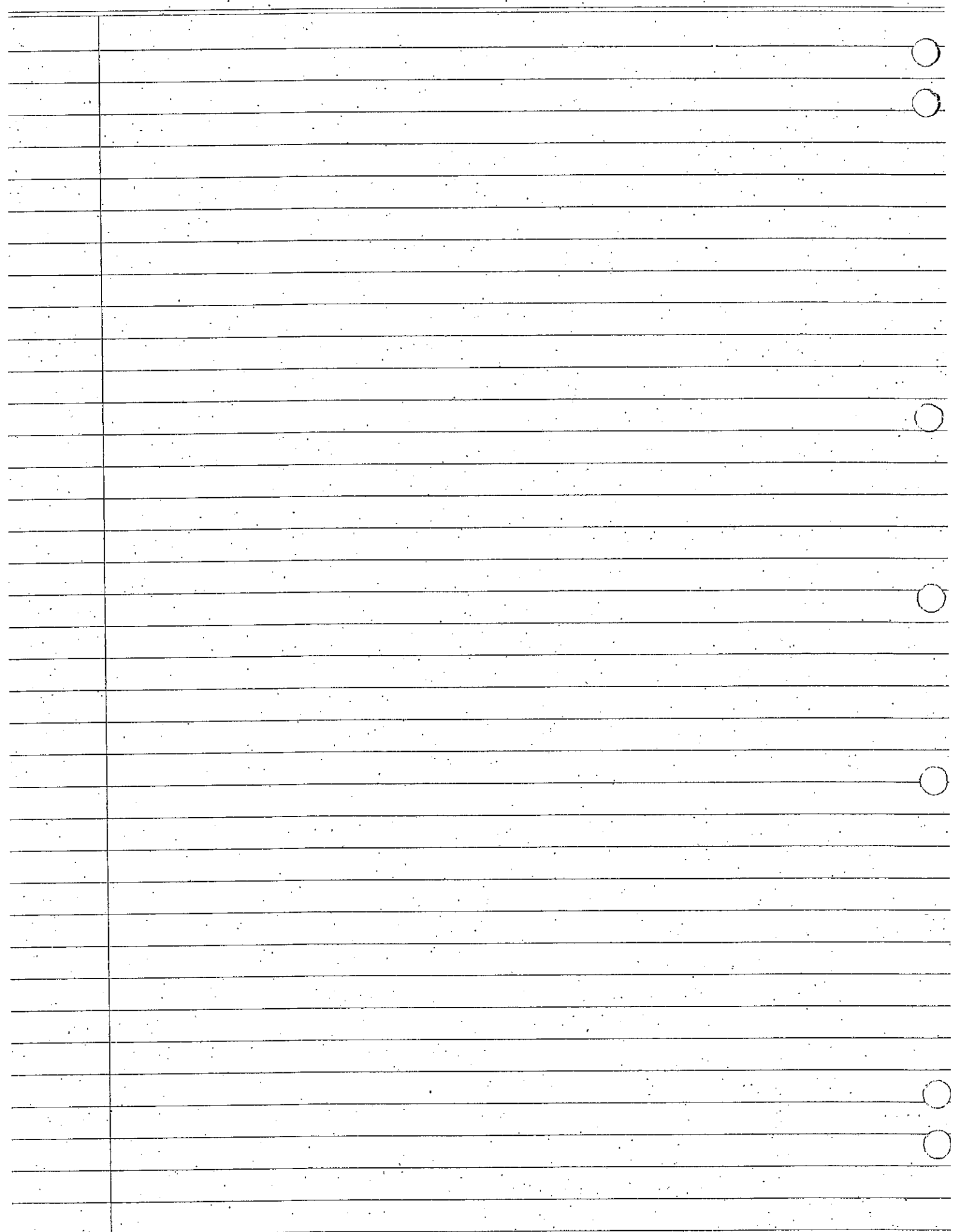
$$V = \pi \cdot \frac{128}{21} \text{ units}^3 \text{ or } \frac{128}{21} \pi \text{ units}^3 \text{ (1 mark)}$$

final
answer)

or correct

V. for their
 V_1 and V_2

i.e. correct.

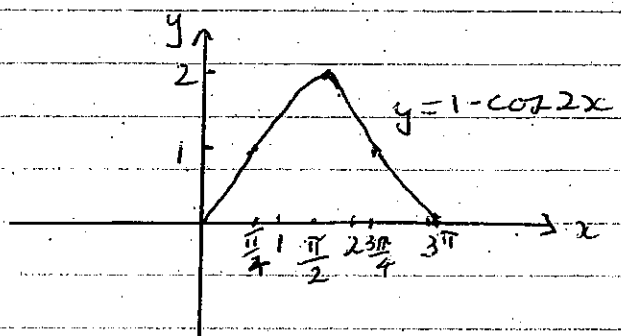


15a $y = 1 - \cos 2x$
for $0 \leq x \leq \pi$

(1 mark for correct shape)

(1 mark for curve between $y=0$ and 2)

(1 mark, clearly shows values for $\frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}$)



b $2 \sin^2 x - 3 \sin x - 2 = 0$

$(2 \sin x + 1)(\sin x - 2) = 0$

$\sin x = -\frac{1}{2}$ or $\sin x = 2$ (1 mark)

↓ no solution

as $y = \sin x$ lies between
-1 and 1

$\therefore x = \frac{7\pi}{6}$ in 3rd & 4th
quadrants

$\therefore x = \frac{7\pi}{6}, \frac{11\pi}{6}$ when $0 \leq x \leq 2\pi$ (1 mark)

15c $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$

$$= \left[\frac{1}{2} \tan 2x \right]_0^{\frac{\pi}{8}}$$

(1 mark)
correct integration

$$= \left(\frac{1}{2} \times \tan \frac{\pi}{4} \right) - \left(\frac{1}{2} \times \tan 0 \right)$$

$$= \frac{1}{2} \times 1 - 0$$

$$= \frac{1}{2} \quad (1 \text{ mark})$$

d i $l = r\theta$

$$l = 5 \times \frac{\pi}{3} = \frac{5\pi}{3} \text{ cm}$$

ii sector Area

$$A_s = \frac{1}{2} r^2 \theta$$

$$A_s = \frac{1}{2} \times 5^2 \times \frac{\pi}{3} = \frac{25\pi}{6} \text{ cm}^2$$

(1 mark)
Area of sector

Triangle ABC Area

$$A_{ABC} = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times 5 \times 5 \times \sin \frac{\pi}{3} = \frac{25}{2} \times \frac{\sqrt{3}}{2} = \frac{25\sqrt{3}}{4} \text{ cm}^2$$

(1 mark)
Area of Δ

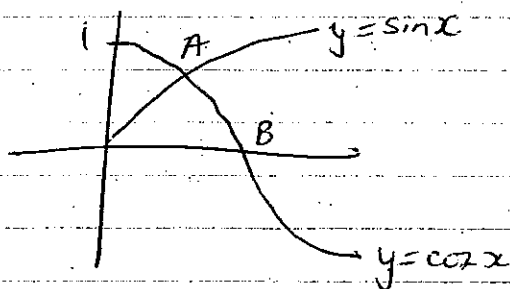
$$\therefore \text{Area} = A_s - A_{ABC}$$

$$= \frac{25\pi}{6} - \frac{25\sqrt{3}}{4}$$

$$= \frac{50\pi}{12} - \frac{75\sqrt{3}}{12} = \frac{25(2\pi - 3\sqrt{3})}{12} \text{ cm}^2$$

(1 mark)
final Area
or ECF with
correct
method.

15e/1 $y = \cos 2x$, $y = \sin x$



$\cos 2 = 0$, when $x = \frac{\pi}{2}$

let $\cos 2x = \sin x$

$\cos x - \sin x = 0$

$\therefore \cos 2x = \sin(90 - x)$

$\sin(90 - x) - \sin x = 0$

$\therefore 90 - x = x$

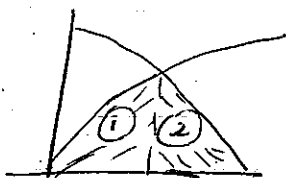
$90 = 2x$

$\therefore x = 45^\circ = \frac{\pi}{4}$

(1 mark working)
that gets A and B
x-values

$\therefore A = \left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$, $B = \left(\frac{\pi}{2}, 0\right)$. (1 mark, final answers)

4



Area = Area ① + Area ②

Area ① = $\int_0^{\frac{\pi}{4}} \sin x \, dx$

$= [-\cos x]_0^{\frac{\pi}{4}}$

$= -\frac{1}{\sqrt{2}} - (-1) = 1 - \frac{1}{\sqrt{2}}$ units²

(1 mark correct method)
split area into 2 areas

Area ② = $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos 2x \, dx$

$= \left[\sin x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$

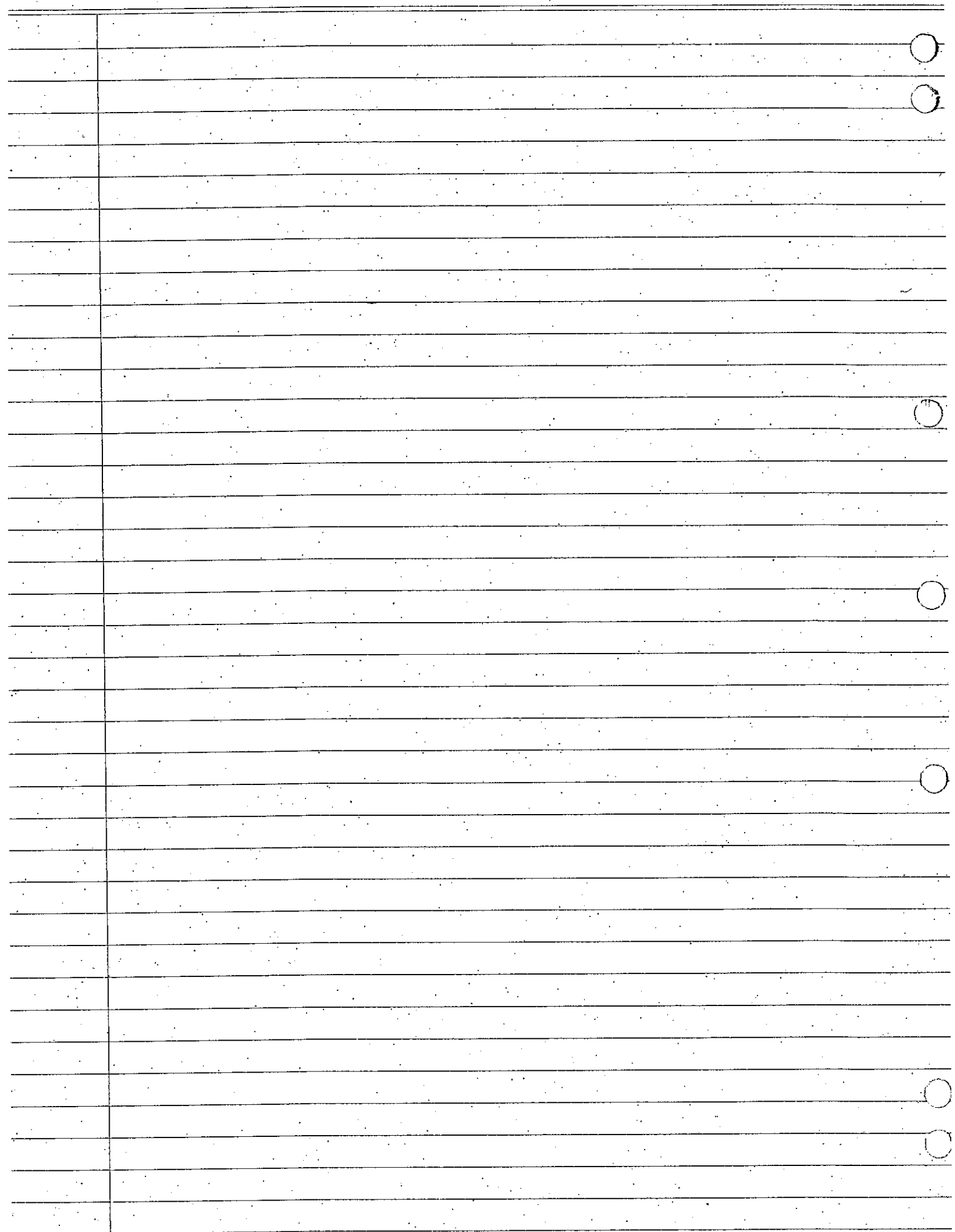
$= 1 - \frac{1}{\sqrt{2}}$

Area = Area ① + Area ②

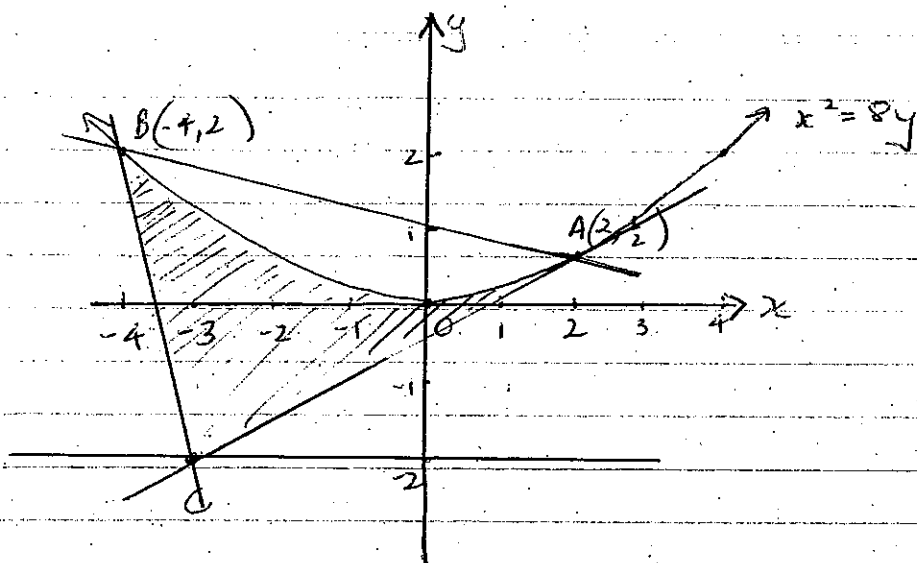
$= 2\left(1 - \frac{1}{\sqrt{2}}\right)$

$= 2 - \sqrt{2}$ units² (1 mark)

correct answer



Q16



a i $x^2 = 8y$

$x^2 = 4ay$ $\therefore a = 2$ $\therefore y = -2$ is directrix

ii $y = \frac{x^2}{8}$

$y' = \frac{x}{4}$

at $x = 2$ $m_t = \frac{1}{2}$

(1 mark for gradient of tangent)

$y - y_1 = m(x - x_1)$

$\left[y - \frac{1}{2} = \frac{1}{2}(x - 2) \right] \times 2$

$2y - 1 = x - 2$

$\therefore x - 2y - 1 = 0$

(1 mark, correctly shows by any method)

iii

let $C = (x, -2)$, C lies on $x - 2y - 1 = 0$

$x - 2y - 1 = 0$

$x_1 - 2(-2) - 1 = 0$

$x_1 + 4 - 1 = 0$

$\therefore x_1 = -3$

$\therefore C = (-3, -2)$

$$\text{IV Area} = \int_{-4}^2 \left(\frac{1-x}{4} - \frac{x^2}{8} \right) dx$$

$$= \left[\frac{x}{4} - \frac{x^2}{8} - \frac{x^3}{24} \right]_{-4}^2 \quad (1 \text{ mark})$$

correct integration.

$$= \left[\frac{2}{4} - \frac{2^2}{8} - \frac{2^3}{24} \right] - \left[\frac{-4}{4} - \frac{(-4)^2}{8} - \frac{(-4)^3}{24} \right]$$

$$= \frac{1}{2} - \frac{1}{2} - \frac{1}{3} - \left(-1 - 2 + \frac{8}{3} \right)$$

$$= 4\frac{1}{2} \text{ units}^2 \quad (1 \text{ mark})$$

final answer.

V Area of ΔCBA

distance AC $A = (2, \frac{1}{2})$, $C = (-3, -2)$

$$d = \sqrt{(2 - (-3))^2 + (\frac{1}{2} - (-2))^2}$$

$$d = \sqrt{5^2 + (\frac{5}{2})^2}$$

$$d = \sqrt{\frac{25 + 25}{4}} = \sqrt{\frac{125}{4}} = \frac{\sqrt{125}}{2} = \frac{5\sqrt{5}}{2} \text{ units}^2$$

perp-distance from $B(-4, 2)$ to $x - 2y - 1 = 0$

$$d_p = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} = \frac{|1x - 4 + -2 \times 2 + -1|}{\sqrt{1^2 + (-2)^2}}$$

$$d_p = \frac{|-4 + -4 + -1|}{\sqrt{5}} = \frac{9}{\sqrt{5}} \text{ units} \quad (1 \text{ mark})$$

perp distance

$$\therefore \text{Area } \Delta CBA = \frac{1}{2} \times \frac{9}{\sqrt{5}} \times \frac{5\sqrt{5}}{2} = \frac{45}{4} \text{ units}^2 \quad (1 \text{ mark})$$

from iv Shaded Area = ΔCBA - Area between AB and curve

$$= \frac{45}{4} - 4\frac{1}{2}$$

$$= 6\frac{3}{4} \text{ units}^2 \quad (1 \text{ mark})$$

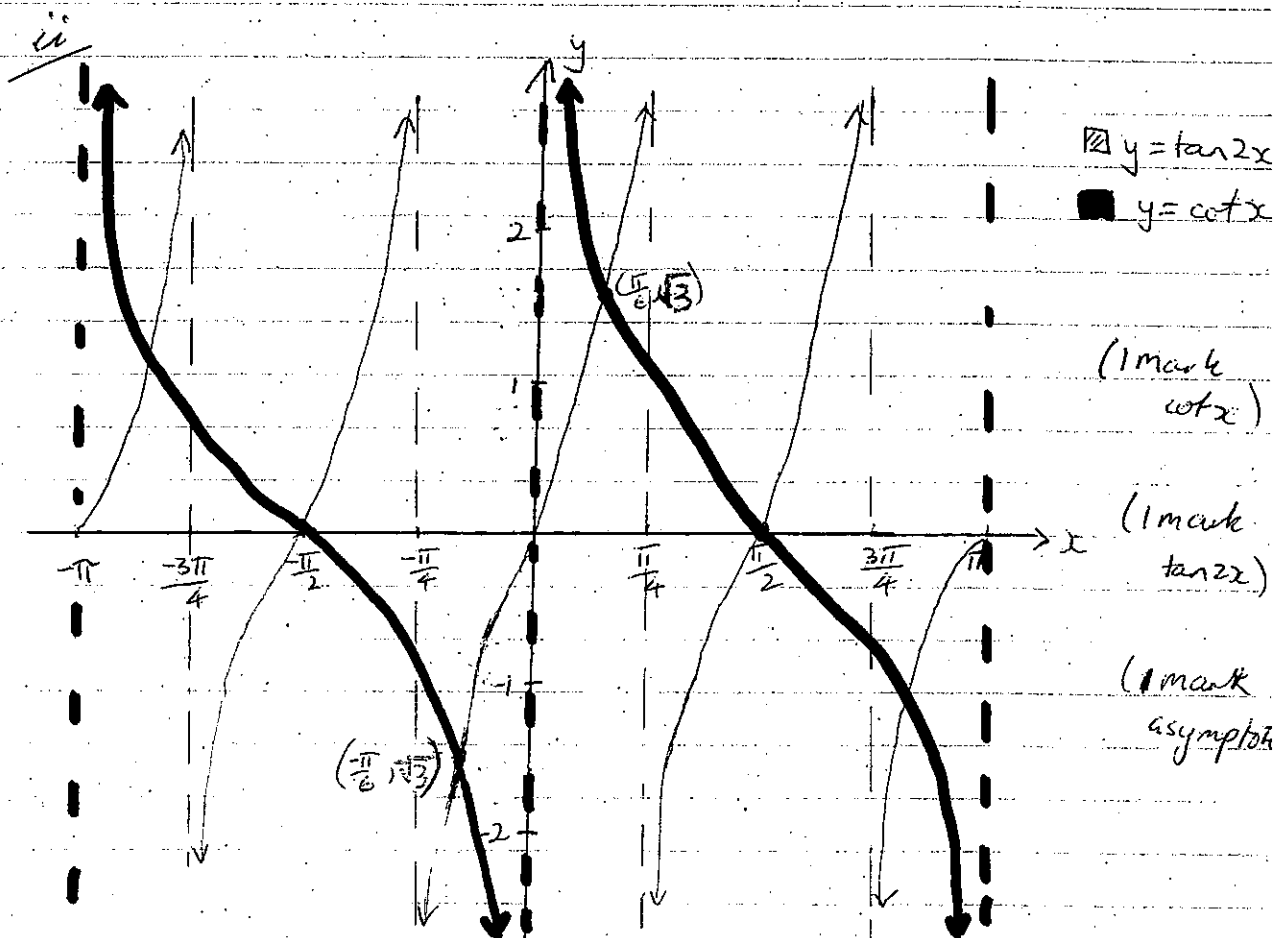
correct answer or ECF

Q16 b i $\cot x = \tan 2x$
 sub $x = \frac{\pi}{6}$

$$\text{LHS} = \cot \frac{\pi}{6} = \sqrt{3}$$

$$\text{RHS} = \tan \frac{\pi}{3} = \sqrt{3}$$

$\therefore x = \frac{\pi}{6}$ is a solution
 (correctly shown by any valid method)



iii $x = \frac{\pi}{6}, \frac{\pi}{2}, -\frac{\pi}{2}, -\frac{\pi}{6}$ from graph.

iv $\frac{\pi}{6} \leq x < \frac{\pi}{4}$ and $-\frac{\pi}{6} \leq x < 0$ and $-\frac{\pi}{2} \leq x < -\frac{\pi}{4}$

